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MOTION PATTERN PREDICTION IN DYNAMIC ENVIRONMENTS

BY

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THESIS

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ABSTRACT

Traditionally, motion planning algorithms have tackled the challenge of navigation in dynamic environments by approximating a robot's configuration space through a graph representation. This involves predicting or computing the trajectories of obstacles and finding feasible paths via a pathfinding algorithm. In our work, we strive to enhance the performance of these subproblems by learning to identify regions critical to dynamic environment navigation. We present a novel methodology for constructing sparse probabilistic roadmaps, a two stage approach that combines a self-supervised learning method for recognizing the topology and geometries indicative of motion patterns in dynamic settings, and a sampling-based strategy for leveraging these learned features. The result is the creation of neural networks capable of predicting the probability of occupancy of a given region and Avoidance Critical Probabilistic Roadmaps (ACPRMs), which leverage these insights to significantly improve navigation performance. ACPRMs have shown remarkable performance, demonstrating up to five orders of magnitude improvement over grid-sampling in multi-agent scenarios and surpassing competitive baselines by up to ten orders of magnitude in multi-query situations.

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CHAPTER 1: INTRODUCTION

1.1 BACKGROUND

In robotics, navigating dynamic environments is a complex task that involves calculating a trajectory for a robot to reach its destination without colliding with moving entities, including people and other robots. This task, known as motion planning, entails finding a feasible and collision-free path within a specific state representation, amidst various static and moving obstacles. A particularly difficult instance of this is Multi-Agent Motion Planning (MAMP), which seeks to compute paths for multiple robots simultaneously, ensuring they avoid collisions with each other. Given the inherent complexity of MAMP, most research in this field focuses on Multi-Agent Pathfinding (MAPF), a discrete space graph search problem that requires accurately representing both the robot and its environment.

A prevalent approach in this domain is Sampling-based Motion Planning (SBMP) [1], exemplified by techniques like Probabilistic Roadmaps (PRMs) [2]. These methods simplify motion planning to a pathfinding problem by forming an implicit graph that maps a robot's configuration space, using samples connected via a collision-checking algorithm [1]. However, a notable challenge arises with conventional sampling strategies in MAMP, such as grid-sampling [3]. These techniques often fail to consider workspace topology and obstacle avoidance, leading to the creation of dense roadmaps that are impractical for navigating tight spaces in multi-agent scenarios. Notably, the primary bottleneck in SBMP arises when computing paths through a roadmap with numerous vertices and edges, all the while managing the conflicts between multiple agents.

To address this, we propose the construction of sparse roadmaps, constructed to accommodate the needs of multiple robots. This improvement expands upon concepts presented in [4]. The key difference is that our approach identifies regions critical to MAMP rather than single-agent motion planning, encompassing both key configurations for static environment trajectories (such as doorways) and those aiding in the construction of broader motion structures necessary for obstacle avoidance. This model employs an adapted version of *betweenness centrality* [5], a graph-theoretic metric traditionally used to gauge the visitation count of a vertex in the shortest paths within a graph. By extending this concept to dynamic settings, we enable agents to gather data necessary for training a model that can discern whether a configuration is useful for avoiding obstacles based on a local occupancy grid. The algorithm we've developed then strategically samples within these pertinent regions, creating what we term a 'learned skeleton' of the workspace. This structure captures the

regions needed for efficient navigation and obstacle avoidance.

The complexity of the problem stems from the presence of other agents or dynamic obstacles, particularly in spatially constrained settings like residential houses or schools. Here, the challenges are twofold: spatial limitations such as doorways and hallways significantly restrict maneuverability, and they also establish unique patterns of interaction between agents [6]. Our solution, Avoidance Critical Probabilistic Roadmaps (ACPRM), demonstrates that configurations often utilized by robots to halt momentarily – indicative of active collision avoidance [7] – can be effectively identified by neural networks. These networks can ascertain whether a given configuration is crucial for avoiding collisions based on local topology and geometry. More generally, our methodology also encompasses the prediction of long-horizon motion patterns. This capability allows agents to 'foresee' and strategically plan for potential encounters, thereby preemptively avoiding collisions and reducing social and path costs. We achieve this through a novel approach of modeling motion pattern prediction as either classification or segmentation over traversability maps, resulting in a task-agnostic feature representation. This representation adeptly captures the intricate relationship between centrality in a graph and the overall topology of the environment.

This advanced approach significantly enhances the ability of agents to navigate complex dynamic environments, particularly in constrained pedestrian spaces like apartments or crowded corridors where conventional navigation systems might struggle. By integrating neural network models with graph-theoretic concepts, our methodology offers an additional representation for tackling some of the multifaceted challenges of multi-agent navigation and social robotics. It holds the potential to improve the interaction dynamics between robots and humans in shared spaces, paving the way for more harmonious coexistence and cooperation in future robotic applications.

1.2 THESIS OUTLINE

Chapter 2: Related Work This chapter provides an overview of the existing research landscape that intersects with the various aspects of this thesis, spanning across robotics and machine learning. It is important to position our work within the broader context of the field, particularly highlighting its connections and distinctions from established methodologies. A significant focus is placed on delineating how our research relates to and expands upon the foundational work of [4]. This chapter serves not only to acknowledge previous contributions but also to contextualize the novel approaches and findings of this thesis.

Chapter 3: Avoidance Critical Probabilistic Roadmaps Adhering to the chronological development of our research, this chapter introduces the concept of Avoidance Critical Probabilistic Roadmaps, which were published at the International Conference on Robotics and Automation in 2021 [8]. This chapter delves into the initial exploratory stages of our research, where we experimented with simple grid-world environments. The primary focus here is on the extension and adaptation of learned probabilistic roadmaps, originally designed for single-query single-agent scenarios, to multi-agent motion planning. This exploration is built upon the theoretical and practical underpinnings established by [4], and it aims to tackle the intricacies involved in adapting these methods to accommodate multiple agents.

Chapter 4: Motion Pattern Prediction in Dynamic Environments This chapter delves into our most recent advancements and efforts in applying the concepts developed in earlier research to real-world environments. This transition necessitated a significant methodological shift, particularly the integration of convolutional neural networks and a departure from our earlier focus on probabilistic roadmaps. The chapter discusses in detail the development and application of machine learning models for the prediction of motion patterns within dynamic environments. This includes an examination of how these models facilitate efficient navigation and social interaction, highlighting the complexities of implementing such systems in real-world settings.

Chapter 5: Conclusions and Future Work The concluding chapter brings together the key findings and contributions of this thesis, offering an analysis of their implications and limitations. This section not only reflects on the achievements of the research but also critically examines areas that warrant further exploration. It outlines potential future directions for continued investigation and opportunities for innovation.

CHAPTER 2: RELATED WORK

2.1 MULTI-AGENT MOTION PLANNING

Multi-agent navigation in robotics presents various challenges, tackled through multiple approaches. Pioneering studies such as those by [9], [10], and [11] have laid the groundwork by proposing the computation of individual agent paths, followed by iterative conflict resolution until all paths are viable. [12] adopts a decentralized approach, assuming uniform collision-avoidance behaviors across all agents. This method is particularly effective when coupled with the prediction of dynamic obstacles' trajectories, enabling pathfinding algorithms like [13] to compute collision-free paths, incorporating the element of time into their search space.

Despite these advancements, most existing solutions operate within discrete state spaces, posing limitations for robots with extensive degrees of freedom or within large environments. These approaches often fall short in supporting heterogeneous robot systems or in providing a comprehensive framework for collision checking and navigation in three-dimensional spaces. Innovations in this field, such as those introduced by [14], extend Multi-Agent Path Finding (MAPF) algorithms into MAMP methodologies using probabilistic roadmaps.

However, methods like [15] and [16], while generating individual roadmaps for each agent and using efficient graph data structures to handle inter-agent conflicts, do not adequately address sampling efficiency in expansive environments with complex topologies. In contrast, [17] and [18] generate unified roadmaps for all agents, primarily focusing on integrating these roadmaps through adaptive sensing and planning under uncertainty in unconstrained or 'open' environments.

Our approach diverges from these methods by constructing a singular, multi-query roadmap designed to be used by all agents. This roadmap aims to simultaneously reduce the complexity of the search and planning processes while enhancing the roadmap's overall reusability and coverage. While [19] proposes a similar strategy, it does not address the fundamental issue of sampling complexity and pathfinding in large graphs, as it requires a dense, pre-existing roadmap as its base. Instead, we tackle this challenge by biasing our sampling process to create roadmaps endowed with the desired attributes for effective multi-agent navigation.

2.2 SAMPLING IN REGIONS CRITICAL TO MOTION PLANNING

In the context of dynamic environment navigation, Spatial Contextual Awareness (SCA) plays a pivotal role. SCA involves understanding the robot's interaction with its surroundings, a critical factor in maneuvering through dynamic spaces. Our focus, particularly with the Avoidance Critical Probabilistic Roadmap (ACPRM), is on explicitly sampling regions critical to Multi-Agent Motion Planning (MAMP).

Various methodologies prioritize sampling near obstacles or within narrow passages ([20], [21], [22]), while others, like [23] and [24], utilize workspace decompositions to guide their sampling process. Recent advancements in the field, such as those made by [25], [26], and [27], have demonstrated the feasibility of learning distributions for sampling purposes.

In contrast to the conventional single-query methods in static environments, our approach emphasizes the development of sparse, multi-query roadmaps in dynamic environments. These roadmaps facilitate efficient obstacle avoidance in large environments characterized by narrow passages. Our strategy, as presented here and in [8], involves constructing learned skeletons, which include configurations within all regions deemed critical for obstacle avoidance, thereby extending the concept of learning sampling distributions to dynamic environment navigation.

2.3 LEARNED CRITICAL PROBABILISTIC ROADMAPS

In the absence of heuristics to pinpoint regions crucial for obstacle avoidance, we adopt a framework akin to the one proposed in [4]. This approach involves solving numerous one-to-all problems across different roadmaps and monitoring the frequency of vertex visits. The derived betweenness centrality values are then used as training labels for a regression neural network, which in turn predicts a normalized 'criticality' score. These scores are pivotal in constructing Critical PRMs, which discern a subset of critical configurations through pre-sampling and aim to connect all subsequent samples to these identified configurations. Our approach, however, diverges in two fundamental ways: firstly, we extend the notion of critical configurations to dynamic environment navigation; and secondly, instead of uniformly pre-sampling to discover a subset of critical configurations, we develop learned skeletons that encompass configurations in all critical regions. This methodology, as shown by [4] and our work, leverages neural networks to successfully identify relevant regions from both grid-world and high-dimensional real-world environment local representations.

2.4 SOCIAL NAVIGATION

Safe navigation amongst humans is a paramount concern for real-world robotic deployment, and it represents one of the most extensively researched areas in robotics. Initial solutions to this challenge treated motion planning and control as separate entities, often resulting in the robot becoming immobilized in scenarios where feasible solutions were not discernible from the predicted pedestrian trajectories ([28], [29]). Modern strategies leverage imitation or reinforcement learning to fuse prediction and planning to either emulate human navigation patterns ([30], [31], [32]) or optimize for an expected reward ([33], [34], [6]). While these methods provide a framework that encapsulates the complexities of social navigation, they are typically confined to the state space regions covered during training, hence needing more scalability and flexibility for life-long learning applications. Our work delves into how machine learning can be utilized to extract pertinent features for both learning and non-learning techniques (such as ACPRM), based on accumulated experience over a robot's lifespan. We posit that models like the one proposed in this paper are adept at learning patterns that facilitate preemptive collision avoidance, life-long learning, and context-dependent behaviors, such as adjusting strategies based on the time of day.

2.5 HUMAN TRAJECTORY PREDICTION

Our proposed approach diverges significantly from existing research in human and traffic modeling. Rather than explicitly predicting the future trajectories of pedestrians, our model estimates a region's occupancy probability based on prior experience. Furthermore, our model's requirement for only a traversability map as an input and its ability to generalize to previously unobserved topologies (e.g., unvisited houses or apartments) represents a significant departure from conventional methodologies.

In navigation, active collision avoidance of unseen agents typically necessitates additional traffic information. Various systems, primarily used for surveillance and monitoring, estimate human activity and flow through stationary sensors like security cameras, tracking pedestrian trajectories ([35], [36], [37], [38]). However, most trajectory prediction in robotics focuses on leveraging information about the recent motion of pedestrians within sensing range to foresee their future positions ([39]). Of particular relevance to our work is the subset of methods that explicitly consider environmental geometry and topology, such as [40], [41], [42], and [43].

Existing methods for motion pattern prediction often struggle to scale in robotic applications due to their reliance on extensive sensor networks and susceptibility to errors in novel

perspectives or scenes ([39]). Our approach differs by abstracting out trajectory prediction and active monitoring, instead relying on a dataset of traversability maps annotated with traffic flow heat maps, generated by either a sensor network or robots. Consequently, we model motion pattern prediction as a probability of occupancy over extended periods, rather than as near-future trajectory prediction.

2.6 IMAGE SEGMENTATION AND CLASSIFICATION

Given that motion patterns often span multiple regions and that the significance of one region is not independent of its surroundings, we approach the prediction of motion patterns as a segmentation/classification problem. Image segmentation, a fundamental problem in computer vision, involves dividing images into pixel clusters that correspond to different segments or objects. In recent years, deep learning has emerged as the leading method for segmentation, owing to its superior performance in various vision tasks ([44]). We employ Fully Convolutional Deep Neural Networks (FCNs) ([45]), characterized by their convolutional layers, use of skip connections for resolution robustness, and ability to produce segmentation maps that match the input size. A notable aspect of FCNs is their loss functions, which aggregate prediction errors for all pixels in the input image, an advantageous feature when the classification of each pixel is interdependent with those around it. For our implementation, we utilize U-Nets ([46]) and fully-connected neural networks, commonly used and straightforward architectures that have demonstrated success in multiple domains.

CHAPTER 3: AVOIDANCE CRITICAL PROBABILISTIC ROADMAPS

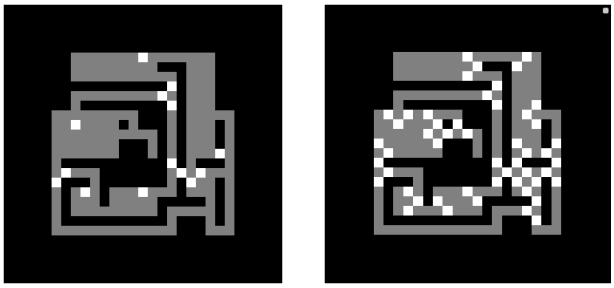
3.1 OVERVIEW

As detailed in Chapter 1, the initial segment of our exploration delves into the concept of Avoidance Critical Probabilistic Roadmaps (ACPRMs) as described in [8]. This particular research endeavor is centered on expanding the methodology introduced in [4], which focused on single-query single-agent probabilistic roadmaps, adapting it to the complexities of the multi-agent setting. This chapter discusses how we approached this extension, considering the limitations of parallel multi-agent simulations available at the time and our prioritization of algorithmic development over environmental realism. Consequently, our initial investigations employed randomly generated grid-world environments to generate training data. This decision aimed to isolate and emphasize the algorithmic challenges inherent in transitioning from a single-agent to a multi-agent context. The simplicity of these environments allowed for a more precise focus on the algorithmic developments and their implications for multi-agent motion planning. However, it's important to note that this approach also had its limitations, primarily regarding realism, some of which we address in Chapter 4.

3.2 PROBLEM STATEMENT

This work aims to solve the multi-agent motion planning problem efficiently by harnessing the capability to identify regions crucial for obstacle avoidance. We focus on three-dimensional planar navigation tailored for omnidirectional robots with rigid bodies, operating within a configuration space denoted as $\mathcal{C}_{space} = \mathbb{R}^2 \times \mathbb{S}$. More precisely, each agent possesses an x, y translational component coupled with a θ rotational component. We define the multi-agent motion planning problem for N such agents as follows: Let $\mathcal{C}_{obst}$ represent the subset of the configuration space occupied by static obstacles, and let $\mathcal{C}_{free} = \mathcal{C}_{space} \setminus \mathcal{C}_{obst}$ define the free space. For each agent i, we denote $s_i \in \mathcal{C}_{free}$ and $g_i \in \mathcal{C}_{free}$ as their respective initial and goal configurations. A path p_i for an agent is considered *collision-free* if it exclusively traverses within $\mathcal{C}_{free}$ and avoids collisions with other agents. Furthermore, a path p_i is deemed *feasible* if it not only meets the criteria of being collision-free but also successfully navigates the i-th agent from its starting point s_i to its intended destination g_i.

Our approach employs sampling-based motion planning techniques, simplifying the motion planning problem into pathfinding. The algorithm constructs an implicit representation of the configuration space through a network of samples connected by local pathways. Each



a) Criticality Data b) Learned Criticality

Figure 3.1: Environment annotated with a) avoidance criticality from 236 random dynamic environment problems and b) a model trained to identify critical regions. White cells are avoidance critical regions, gray cells are free space, and black cells are static obstacles.

connection is validated for safety and viability through a collision-checking algorithm [1]. Given a roadmap that effectively connects each agent's starting point s_i to their goal g_i for all i, we compute feasible paths p_i for every agent by employing a Multi-Agent Pathfinding (MAPF) algorithm. This algorithm ensures that each agent's path is individually feasible, avoiding any potential inter-agent conflicts or collisions.

3.3 DEVELOPING SPATIAL CONTEXTUAL AWARENESS

3.3.1 Identifying Avoidance Critical Regions

One of our goals is to define a methodology for identifying configurations critical to dynamic environment navigation. In this work, we extend the concept of betweenness centrality to dynamic environments by computing many one-to-one problems. For simplicity, we consider the case where the agent has prior knowledge of the moving obstacles' trajectories. However, this assumption is not imperative; an agent may actively predict the dynamic obstacles' trajectories during training or test time. In practice, we found that utilizing betweenness centrality (i.e., tracking the number of time steps an agent occupied a specific configuration) resulted in the obfuscation of regions critical to dynamic environment navigation by those critical to static-environment navigation. Therefore, our metric only increases when an agent is waiting in place and the dynamic obstacles are within a user-defined distance. This subtle contextual cue (the waiting in place event) is unique to dynamic environment navigation and implies that the agent is actively avoiding a collision. This metric, the amount of time an agent chose to wait at a configuration, is what we refer to as *avoidance criticality*. However, the nature of dynamic environment navigation leads to

Figure 3.2: Randomly chosen avoidance critical occupancy grids. From darkest to lightest the cells represent: static obstacles, free space, the agents' region of origin within the occupancy grid, the avoidance critical region (center of the grid), and the agent's goal region within the occupancy grid.

this being an ill-defined characterization of our method. There are two additional criteria:

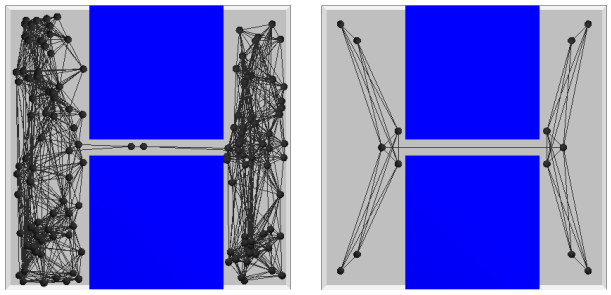
- Avoidance criticality may be computed in environments with any number of obstacles, each with an arbitrary trajectory, size, speed, or behavior. In our data-generation procedure, we introduce three dynamic obstacles of the same size as the agent per navigation task, each of which takes the shortest path to get to its goal and moves with constant speed. One of the obstacles travels from the goal to the start of the agent's motion task, requiring the agent to avoid at least one collision. The other two obstacles are each assigned a random start and goal.

- Using pathfinding algorithms that provide no guarantees on the quality of the shortest path they find, such as A* [47], can result in unpredictable obstacle avoidance behavior. An agent waiting for a narrow passage to clear may wait at any configuration or compute a suboptimal trajectory to "waste time". For this reason, we use safe interval path planning (SIPP) [13], which guarantees that the agent arrives at each configuration in the solution trajectory at the earliest time possible.

Fig. 3.1a shows an example of a randomly generated environment annotated with avoidance criticality.

3.3.2 Learning to Recognize Avoidance Critical Regions

We use avoidance criticality to get supervision signal for a neural network model. While betweenness centrality typically increases as it gets closer to local maxima, avoidance criticality does not display the same property. As seen in Fig. 3.1a, it instead consists of clusters of critical regions, with some being more critical than others due to workspace topology. As a result, we frame our problem as a classification task that bins avoidance criticality scores into a user-defined number of avoidance criticality levels. Here, we concentrate on the binary



a) PRM b) ACPRM

Figure 3.3: PRM (a) and ACPRM (b) used to compute motion plans for 4 agents in a narrow passage environment.

classification setting. In other words, although avoidance criticality refers to the number of timesteps agents spent waiting in a region, we choose to use our model to predict whether or not a region is likely to be used to wait in place for any amount of time.

Dataset Creation We use self-supervision to train a neural network h_θ to determine whether a region is avoidance critical or not. Section 3.3.1 outlines the procedure for annotating an arbitrary environment. However, to generalize to a wide range of workspace topologies, the training data for h_θ is collected from various environments. We generate a dataset Y of criticality label y and occupancy grid $g(x)$ pairs. Where $y \in [0, 1]$ indicates whether or not a sample is avoidance critical, $x \in \mathcal{C}_{free}$ is a configuration, and $g(x)$ is the occupancy grid representation of the workspace around x. Fig. 3.2 shows randomly selected avoidance critical samples from the training data. In the interest of planar navigation in human environments, we use a random environment generator that constructs structured environments that consist of grid-aligned rooms and hallways of diverse shapes and sizes [48]. Figs. 3.1a-3.1b depict one such environment.

Learning Using Y, we train h_θ. Which outputs $\hat{y} \in [0, 1]$, an indicator of whether or not the model believes a given occupancy grid contains an avoidance critical region in its center. During training, we minimize cross-entropy loss to learn θ and use Stochastic Gradient Descent (SGD) with momentum to optimize. Namely, we minimize:

$$\mathcal{L}(\theta) = -y \log(h_\theta(g(x))) - (1 - y) \log(1 - h_\theta(g(x))) \tag{3.1}$$

Fig. 3.1b shows an example of a randomly generated environment annotated by h_θ through a sliding window procedure.

Algorithm 3.1: ACPRM

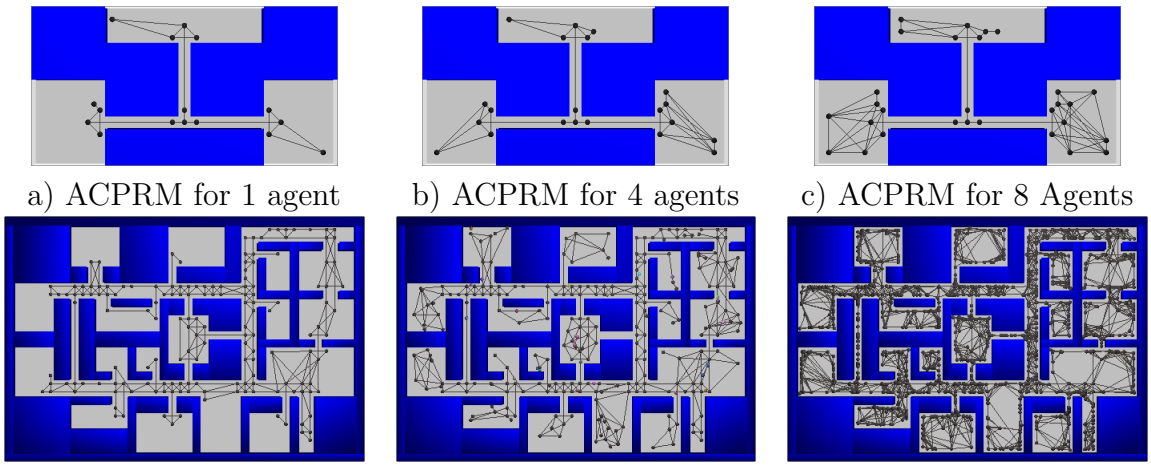
Input: MAMP Problem $(s_i,\ g_i\ i \in \{0, .., N\}, h_\theta, k_a, k_p, PF)$

1: Discretize environment and generate occupancy grid inputs for h_θ
2: Identify avoidance critical regions using h_θ
3: Sample in avoidance critical regions
4: Connect critical samples to k_a nearest-neighbors
5: Sample uniformly in $\mathcal{C}_{space}$
6: Connect non-critical samples to k_p nearest-neighbors
7: Continue sampling and connecting until pathfinding algorithm, PF, can compute feasible paths for each agent

Implementation h_θ is a fully connected neural network with 7 hidden layers, each with a ReLU activation function and a subsequent dropout layer set to 0.1. We use SGD with a learning rate of 0.01 and a momentum of 0.9 to optimize. The training data is collected from 450 randomly generated environments. For each region, we store its avoidance criticality score and an 11×11 occupancy grid with the critical region in its center. Our dataset contains 70K critical and 70K non-critical examples.

3.4 AVOIDANCE CRITICAL PROBABILISTIC ROADMAPS

This section presents a hierarchical SBMP method we call Avoidance Critical Probabilistic Roadmap (ACPRM). Algorithm 3.1 outlines the entire procedure. First, we compute a discretized representation of the environment to generate the occupancy grid inputs for h_θ (Line 1). Then, ACPRMs build a learned avoidance critical skeleton by sampling in all regions h_θ deems critical through a sliding window procedure. Specifically, ACPRMs sample in avoidance critical regions (Line 2-3) and then connect all samples to their k_a nearest neighbors (Line 4). An example of a learned skeleton is illustrated in Fig. 3.4d. Such skeletons provide motion structures and samples critical to the narrow passage problem. Although any sampling strategy may be used after generating the skeleton, we use a PRM scheme [2] that uniformly samples configurations (Line 5) and connects each sample to its k_p nearest neighbors (Line 6), which may or may not be part of the learned skeleton. After constructing the learned skeleton, ACPRMs iteratively connect each agent's start and goal configurations to the roadmap, sampling uniformly if necessary. Figs. 3.4a-3.4c depict the evolution of an ACPRM as the number of agents increases. Once the roadmap connects all of the start and goal configurations, ACPRM uses an arbitrary multi-agent pathfinding algorithm to find feasible paths for each agent (Line 7). Fig. 3.3 shows a qualitative comparison of two multi-agent roadmaps, where the only difference in the sampling procedure



a) ACPRM for 1 agent b) ACPRM for 4 agents c) ACPRM for 8 Agents

d) Learned Skeleton e) ACPRM f) MAPRM

Figure 3.4: Relevant roadmaps from the three-way junction and building environments.

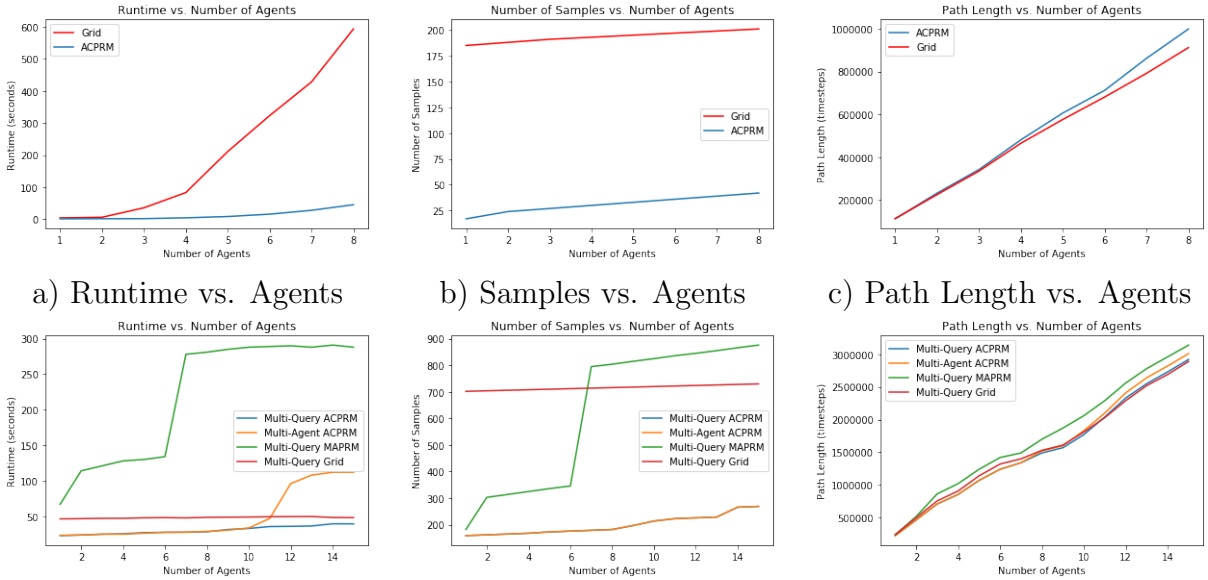
is the construction of the learned skeleton before uniform sampling.

3.5 EXPERIMENTAL RESULTS

3.5.1 Pathfinding in ACPRMs

While ACPRMs will find a solution as the number of samples and connections (k_p) approach infinity, Algorithm 3.1 assumes access to a pathfinding algorithm that can track inter-agent collisions in the roadmap at an arbitrary resolution. In our implementation, we use a custom-made decentralized pathfinding algorithm based on SIPP [13] that allows robots to travel through edges but only wait at vertices. While our implementation is not guaranteed to find the globally optimal solution to the multi-agent motion planning problem, it finds the shortest feasible path for each agent given the obstacles already in the environment.

We chose to implement a custom pathfinding algorithm because existing pathfinding algorithms for obstacle avoidance only scale well in discrete environments (i.e., planar graphs). Multi-agent pathfinding in roadmaps is not trivial as roadmaps have properties such as redundant edges (e.g., three or more vertices that lie on a line all connected to one another) and vertices that collide with one another. Since these properties vastly increase the search and planning space, developing multi-agent pathfinding algorithms that perform well on roadmaps is an open research area.



a) Runtime vs. Agents b) Samples vs. Agents c) Path Length vs. Agents

d) Runtime vs. Agents e) Samples vs. Agents f) Path Length vs. Agents

Figure 3.5: Plots comparing grid-sampling and ACPRM in the three-way junction environment (a-c) and grid-sampling, ACPRM, and MAPRM in the building environment (d-f).

3.5.2 Motion Planning with ACPRMs

In this section, we seek solutions to the difficult problem of constructing a multi-query roadmap that scales well in large environments with narrow passages. Therefore, many standard baselines that sample uniformly [2] or bias sampling towards obstacles [20] could not find solutions without exceeding our computation time limit. Thus, we compare the performance of ACPRM with that of a grid-sampling baseline. Additionally, since ACPRMs should be useful for an arbitrary number of agents, we also compare its performance on multi-query single-agent motion planning against that of grid-sampling and Medial-Axis PRM [49], a method that retracts samples towards the medial-axis of $\mathcal{C}_{free}$ to find narrow passages and improve coverage of the environment.

3.5.3 Three-Way Junction Multi-Agent Problem

For our first multi-agent motion planning problem, we plan in the three-way junction environment shown in Figs. 3.4a-3.4c. This type of environment is known to be difficult for baselines and accentuates some of the features of ACPRMs. As depicted in Fig. 3.5a, ACPRM outperforms grid-sampling and has runtime improvement that increases as the number of agents increases. After eight agents, the uniform grid-sampling baseline exceeded our computation time limit of ten minutes. On the final eight-agent problem, ACPRM found

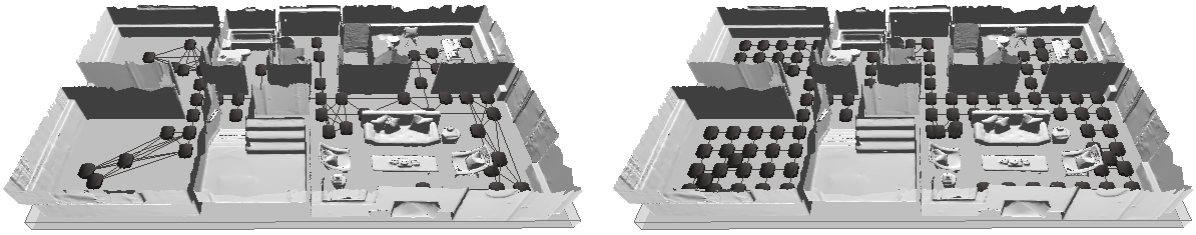
a solution over six times faster. Moreover, Figures 3.5b and 3.5c show that ACPRM finds only marginally longer paths with over eight times fewer samples. Both methods perform similarly when the number of agents is small, which illustrates that the improvements stem from the considerably smaller search space. Figs. 3.4a-3.4c depict the evolution of the ACPRM as the number of agents increases. In Fig. 3.4a, the learned skeleton is visible. It includes configurations critical for motion planning in narrow passages (e.g., entrances and a sample at the junction itself) and emergent motion structures that allow agents to take turns going through the narrow passages efficiently (triangle-shaped subgraphs at entrances and samples around the center of the junction).

3.5.4 Multi-Query Building Problem

To demonstrate ACPRM's scalability and ability to generalize to environments unlike those in the training data, we deploy ACPRM on an environment based on an existing building (See Figs. 3.4d-3.4e). We conduct two types of experiments. The first set seeks to study the use of ACPRM as a multi-agent roadmap. In contrast, the second evaluates ACPRM's coverage of the environment and its merit as a multi-query roadmap for single-agent motion planning. All experiments solve the same 15 randomly generated motion planning problems.

ACPRM for Multi-Agent Motion Planning Fig. 3.5d plots the performance of an ACPRM used for MAMP (Multi-Agent ACPRM) and that of an ACPRM used to solve multiple queries (Multi-Query ACPRM) in the building environment. Both create the same roadmap and, therefore, have the same sampling complexity (Fig. 3.5e). However, after introducing ten agents to the environment, Multi-Query ACPRM begins to outperform Multi-Agent ACPRM by up to three orders of magnitude. These findings are expected, as planning for multiple agents requires adding the time dimension to the search space. Nonetheless, they shed some key insights. First, they illustrate the cost of pathfinding and the importance of keeping the search space small. Since the roadmaps are identical, the additional computation time stems from multi-agent pathfinding. Second, the increase in the total path length (Fig. 3.5f) succinctly shows that collision avoidance in this environment requires agents to wait or find paths longer than those found in the multi-query setting.

ACPRM for Single-Agent Motion Planning We also compare ACPRM, grid-sampling, and MAPRM in the single-agent multi-query setting. Fig. 3.5d shows that ACPRM outperforms both baselines. ACPRM finds solutions up to ten times faster than MAPRM and up



a) ACPRM b) Grid-Sampling

Figure 3.6: Roadmaps in Gibson Environment

to two times faster than grid-sampling. These findings and the inability of other baselines to find solutions, substantiate the merits of using neural networks to bias sampling and construct sparse roadmaps. It is computationally inexpensive to annotate large environments. On the other hand, methods like MAPRM use sampling procedures that do not scale well in large and complex environments. Furthermore, as seen in Figs. 3.5e-3.5f, ACPRM uses up to three times fewer samples than MAPRM and grid-sampling but still finds shorter and roughly equivalent paths, respectively. Figs. 3.4d-3.4e, depict the learned skeleton and the roadmaps generated by both methods.

3.5.5 Real-World Environment

Lastly, we deploy an ACPRM (Fig. 3.6a) and the grid-sampling baseline (Fig. 3.6b) in an environment from the Gibson Database of 3D Spaces [50], a dataset generated by scanning indoor environments. Additionally, we use a model for the omnidirectional Clearpath Ridgeback mobile base. The environment is sufficiently small for grid-sampling to find a solution for eight agents moving in the environment simultaneously in six seconds. ACPRM finds a solution in three seconds and produces marginally shorter paths. In these experiments, both grid-sampling and ACPRM are deterministic, as they cover workspace such that there is no need for uniform sampling, agents need only connect their start and goal configurations to the roadmap. The random generation of starts and goals may make the problem infeasible or trivial, so we present the results of the first randomly generated problem that required obstacle avoidance. This experiment shows that if the training data is collected in appropriate environments and the discretization is adequate, ACPRMs are useful for motion planning in the real-world.

3.6 CONCLUSION

This chapter outlines an efficient method for identifying and leveraging regions crucial to obstacle avoidance in multi-agent motion planning. Central to our approach is using an agent's behavior—waiting for obstacles to pass—as a contextual indicator of the significance of a configuration within the context of obstacle avoidance. This behavior guides the training of a neural network to pinpoint critical regions for dynamic environment navigation.

In this chapter, we present Avoidance Critical Probabilistic Roadmaps (ACPRMs), a significant advancement in motion planning, which utilizes a trained neural network to construct sparse probabilistic roadmaps. These roadmaps are adept at supporting multiple agents in constrained spaces, effectively addressing the challenges of traditional motion planning methods in managing search spaces efficiently. ACPRMs enhance multi-agent motion planning by significantly improving agents' capacity for efficient obstacle avoidance through the identification and integration of critical regions, markedly reducing the search space needed for path planning to boost computational efficiency in complex scenarios, and expanding the roadmap coverage to create a versatile navigational framework adaptable to large environments.

Overall, the advancements presented in this chapter mark a significant contribution to multi-agent motion planning. By integrating neural network models with the principles of probabilistic roadmaps, ACPRMs offer a practical solution to the multifaceted challenges of navigating dynamic environments . This approach enhances the performance and efficiency of motion planning algorithms and opens up new possibilities for more complex and nuanced multi-agent interactions in various settings.

CHAPTER 4: MOTION PATTERN PREDICTION IN REAL-WORLD ENVIRONMENTS

4.1 OVERVIEW

After the initial publication on ACPRMs, our work shifted towards a more in-depth analysis of criticality and visitation counts in dynamic environment navigation. We primarily focused on the theoretical underpinnings of occupancy probability, more advanced simulations, and neural networks for extracting motion patterns from real-world environments. This transition was facilitated by the availability of crowd simulation in real-world apartments through the iGibson challenge at the Conference on Computer Vision and Pattern Recognition 2021. This development departed from a primary focus on multi-agent and sampling-based motion planning, repositioning these methods as applications within a broader context.

Additionally, while initially employing waiting in place as a contextual cue for obstacle avoidance, our research evolved to pursue a more comprehensive understanding of long-term motion pattern prediction. We reoriented the problem to focus on estimating occupancy probability, particularly regarding proximity between agents, as it is a standard metric in social and dynamic environment navigation. This metric serves as a safety indicator and a means to assess method performance, where collisions incur the highest penalties.

Our exploration concentrated on a fundamental instance of this problem: an agent navigating an environment with limited knowledge of other agents' presence or locations, relying solely on access to a map. This application is an enhancement of global planning, where the agent aims to determine the optimal path to proactively avoid collisions. This shift in focus underscores our commitment to developing robust, real-world, applicable solutions in the field of navigation, contributing to a more nuanced understanding of agent interaction and navigation in dynamic spaces.

4.2 PROBLEM STATEMENT

In order to best express and understand the most formal application of our model to social navigation, we use a general formulation for social navigation from [51]. However, we express the optimization in terms of the expected cost. Let $\mathcal{Q}$ be a given planar configuration space for the robot of interest. Given some configuration $c \in \mathcal{Q}$, we denote by $A_R(c)$ the area the robot takes at that position in workspace. At some time $t \geq 0$, the agent has to navigate amongst a set of $\bar{n}_p \geq 1$ pedestrians. Given some configuration $c \in \mathcal{Q}$ for pedestrian $j \in [\bar{n}_p]$, we denote by $A_j(c)$ the volume it takes at that position in workspace. The agent and

pedestrians intend to reach their destinations g_R and $(\bar{g}_j)_{j\in[\bar{n}_p]}$ respectively while avoiding collisions with the a set of static obstacles $\mathcal{C}_{obst}$ and abiding to social norms (e.g., respecting the personal space of others). Let $\mathcal{T} = \{\tau : [0,1] \to \mathcal{Q} \mid \tau(0) = q_R, \tau(1) = g\}$ be the set of all possible trajectories over the configuration space (parameterized as a curve) that start at point $q_R \in Q$ and finished at the goal point $g \in \mathcal{Q}$. At planning time, the agent generates an initial intended trajectory, or global plan, $\tau^*_{\text{ini}} \in \mathcal{T}$ by solving the following optimization problem:

$$\tau^*_{\text{ini}} = arg\,min_{\tau_{\text{ini}}\in\mathcal{T}} \;\; \mathbb{E}_{\tau_f,\overline{\mathcal{T}}\sim\phi(\tau_{\text{ini}},\mathcal{E})}\left[c(\tau_f) + \lambda c^s(\tau_f,\overline{\mathcal{T}})\right] \tag{4.1}$$

$$s.t. \;\; A_R(\tau_i(t)) \notin \mathcal{C}_{obs} \forall t \in [0,1] \tag{4.2}$$

$$\tau_{\text{ini}}(0) = \tau_f(0) = q_R \tag{4.3}$$

$$\tau_{\text{ini}}(1) = \tau_f(1) = g_R \tag{4.4}$$

where $\phi(\tau_{\text{ini}}, \mathcal{E})$ is a simulation of potential navigation episodes given the initial trajectory τ_{ini} and the environmental context $\mathcal{E}$ (e.g., environment geometry/topology, time of day, semantics, etc.). This simulation is stochastic due to the stochastic nature of the pedestrian trajectories. Since the initial trajectory may require additional replanning due to the presence of an arbitrary number of pedestrians with stochastic trajectories, the simulation must compute multiple final trajectories $\tau_f \in \mathcal{T}$ and pedestrian trajectories $\overline{\mathcal{T}} = (\bar{\tau}_1, \ldots, \bar{\tau}_{n_p})$, where $\bar{\tau}_i : [0,1] \to \mathcal{Q}$, $i \in [\bar{n}_p]$. For example, assume the agent intends to take the shortest path τ_{ini} to go to the kitchen. In some scenarios the agent may encounter pedestrians with trajectories $\overline{\mathcal{T}}$ in a narrow passage, resulting in a final trajectory τ_f that captures the additional motions the agent took to backtrack and wait for the narrow passage to be clear of pedestrians. Since there are quasi-infinite potential scenarios, the agent must consider a large number of potential scenarios when choosing its initial path τ_{ini}. In order to quantify the cost of its trajectory the agent must consider $c : \mathcal{T} \to \mathbb{R}$, a cost corresponding to the nature of the path (e.g., how long it took the robot to traverse it) and $c^s : \mathcal{T}^{1+n_p} \to \mathbb{R}$, a λ-weighted cost that captures social considerations and takes into account predictions about the future behavior of others $\overline{\mathcal{T}} = (\bar{\tau}_1, \ldots, \bar{\tau}_{n_p})$. In general, the robot does not have access to the navigation costs of the pedestrians, which includes $\bar{c}_j$, $\bar{c}^s_j$, trajectory $\bar{\tau}_j$, or weight λ of the pedestrians, though each is computing a similar optimization independently. In this work, we refer to this type of navigation as social navigation.

The objective above captures a general form of social navigation. However, it is intractable because the social costs are not strictly defined. Therefore, we focus on the cost of proximity. We model the social cost as the average of the distances to the pedestrians that are within

a distance threshold δ through a zero mean Gaussian density, a common formulation for social costs ([52]). In other words, the social costs are exponentially larger the closer the pedestrian is to the robot. We sum over the Gaussian density from the beginning to the end of the discretized final trajectories sampled. Let $\hat{Q}$ be a discretization of the configuration space. Then, we can define the discrete set $\mathcal{B}_\delta(p) := \{x \in \mathcal{Q} \mid \|x - p\| \leq \delta\} \cap \hat{Q}$. We also define $\hat{\mathcal{T}}$ the set of trajectories taking values in $\hat{Q}$ that start in $q_r \in \hat{Q}$ and finish in $g \in \hat{Q}$ and $\hat{\tau} \in \hat{\mathcal{T}}$, the set of elements in $\hat{Q}$ that are occupied by the trajectory τ. Finally, we discretize the time and assume the robot agent and pedestrians move in a total time interval of $N + 1$ steps, from time step 0 and N.

$$\hat{\tau}^*_{\text{ini}} = arg\,min_{\hat{\tau}_{\text{ini}} \in \mathcal{T}} \;\; \mathbb{E}_{\tau_f \sim \phi(\hat{\tau}_{\text{ini}}, \mathcal{E})} \left[c(\tau_f) + \sum_{n=0}^{N} \frac{1}{|\bar{\Gamma}_\delta(\hat{\tau}_f(n), n)|} \sum_{x \in \bar{\Gamma}_\delta(\hat{\tau}_f(n), n)} e^{-\frac{\|\hat{\tau}_f(n) - x\|^2}{2\sigma^2}} \right] \tag{4.5}$$

where $\bar{\Gamma}_\delta(p, n) = \mathcal{B}_\delta(p) \cap \{q \mid q \in \{\bar{\tau}_j(n)\}_{j \in [\bar{n}_p]}\}$.

Rather than simulate the set of trajectories $\overline{\mathcal{T}}$ given $\mathcal{E}$ we propose to model the probability of finding a pedestrian at any location $x \in \hat{Q}$ (independent of time) with a model $h_\theta(x)$. As discussed in 1, this is the focus of our work. Our goal is to estimate an appropriate h_θ or similar representation that can be used to estimate the social costs relative to other options. Given such a model, it is possible we propose to estimate the social costs with the following expression:

$$\hat{\tau}^*_{\text{ini}} = arg\,min_{\hat{\tau}_{\text{ini}} \in \mathcal{T}} \;\; \mathbb{E}_{\tau_f \sim \phi(\hat{\tau}_{\text{ini}}, \mathcal{E})} \left[c(\tau_f) + \sum_{n=0}^{N} \frac{1}{|\mathcal{B}_\delta(\hat{\tau}_f(n))|} \sum_{x \in \mathcal{B}_\delta(\hat{\tau}_f(n))} h_\theta(x) e^{-\frac{\|\hat{\tau}_f(n) - x\|^2}{2\sigma^2}} \right] \tag{4.6}$$

4.3 METHODOLOGY

4.3.1 Data Generation

The goal of our model is to learn to predict which regions of the map are most frequently visited by pedestrians from experience. We generate our training data by tracking the criticality of single and multiple agents over multiple episodes. Our multi-agent simulation uses reciprocal collision avoidance (ORCA) [53], a commonly used crowd simulation algorithm for social navigation simulations, to simulate trajectories in realistic reconstructions of apartments from the iGibson dataset [54]. We generate relevant ground-truth annota-

a) Map b) Simulated Data c) Predictions

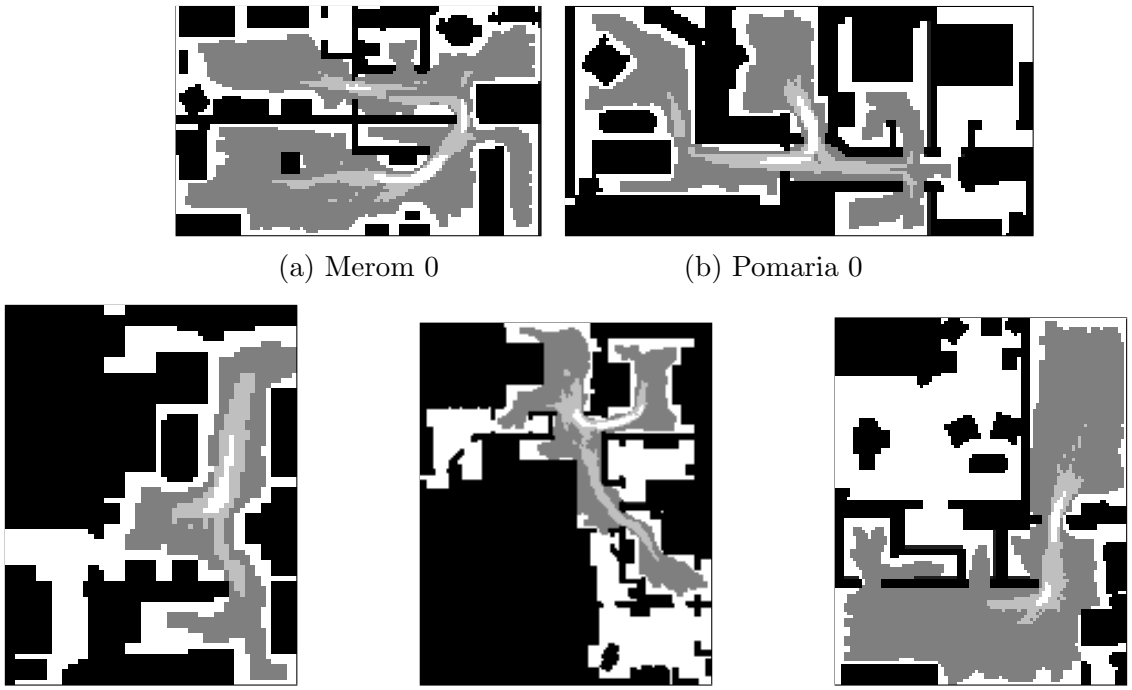
Figure 4.1: Merom 0 environment a) map, b) heat map from hundreds of simulated pedestrian trajectories and c) motion pattern predictions using our approach. As seen, the predictions capture the narrow passages and frequently visited regions of the environment.

tions by simulating multiple episodes and tracking the area swept by the pedestrians during execution. Once the visit counts per pixel are collected, we normalize and bin them into three classes, effectively constructing a segmentation mask where each class corresponds to the "level" of criticality of the region based on the uniform quantization. While three is an arbitrary hyper-parameter, we chose it since we observed that there are roughly three clusters in the motion patterns: chokepoints/narrow passages, regions frequently visited by pedestrians given their proximity to chokepoints and narrow passages, and open spaces. We use this intuition for the qualitative analysis of our results and propose it as a guideline for human-generated labels in the future. Fig. 4.1b shows the segmentation labels for one of the training environments (brighter colors depict more visited regions). This procedure is computed in eight environments, five of which are used for training (shown in Fig. 4.2) and three for evaluation.

The dataset Y consists of segmentation mask labels y and unannotated occupancy grids x. Where $y^i_{p,c} \in [0,1]$ for every pixel p in the i-th grid depending on whether it is labeled as class c or not. There are two special classes, 0 and -1 for pixels in $\mathcal{C}_{obst}$ and unlabeled free space ($\mathcal{C}_{free}$), respectively. These classes are necessary for calculating the loss and weights. The input features x are occupancy grids of size $(n \times n)$, where $x^i_p \in [0,1]$ for every pixel p in the i-th grid depending on whether the location is an obstacle or free space. We extract grids with overlaps from the environments and rotate/mirror them to create eight unique transformations of each example, discarding examples with too few labeled pixels.

4.3.2 Segmentation U-Net

We use a U-Net [46] based implementation, an FCN that downsamples, upsamples, and skips connections to learn features at various resolutions. Using Y, we train h_θ. Which outputs $\hat{y} \in \mathbb{R}^{n \times n}$, a matrix consisting of the criticality predictions for each pixel. During training, we minimize pixel-wise cross-entropy loss and optimize with Adam [55]. Namely,



(a) Merom 0 (b) Pomaria 0

(c) Rs (d) Beechwood 1 (e) Ihlen 0

Figure 4.2: Segmentation masks used as training data.

we minimize:

$$\mathcal{L}(\theta) = -\sum_{p \in y} \sum_{c=1}^{C} w_c (\log \hat{y}_{p,c}) y_{p,c} \tag{4.7}$$

Where p is the pixel position in y and w_c is a weight that depends on the class. Since the training data is imbalanced, we use the inverse of the frequency of each class as its weight. We also have many unlabeled pixels, so we assume that the unlabeled pixels follow the same class distribution as the labeled critical pixels and add them to the labeled class counts to compute the weights. Without increasing the counts of critical pixels, the weights would practically ignore the obstacle class due to the considerably larger number of obstacle pixels relative to labeled free-space pixels.

During training, we set the weight of unlabeled pixels in $\mathcal{C}_{free}$ to 0. Therefore, the optimization does not include unlabeled free space in the loss and prevents the network from predicting pixels without a criticality value. We found that the network benefits from predicting pixels in $\mathcal{C}_{obst}$, effectively learning to predict where obstacles are despite the obstacles being perfectly specified in the input. Notably, the model sometimes predicts obstacles in $\mathcal{C}_{free}$ due to it not being penalized for doing so over unlabeled $\mathcal{C}_{free}$. We assign an additional criticality level to those predictions to avoid creating misspecified traversability maps. The nature of the problem raises interesting research questions outside of the scope of this work.

In the future, we plan to further explore model prediction over topology, where one may not want to predict $\mathcal{C}_{obst}$ nor treat $\mathcal{C}_{obst}$ and unlabeled $\mathcal{C}_{space}$ in the same way in the loss function. Below are details on our specific U-Net implementation and training.

- Input: 64^2 pixel local image (6.4^2 meters)
- Architecture: $32^2 \times 3^2$ conv - max pooling - $64^2 \times 3^2$ conv - max pooling - $128^2 \times 3^2$ conv - max pooling - $128^2 \times 3^2$ conv - upsampling - $64^2 \times 3^2$ conv - upsampling - $32^2 \times 3^2$ - (with residuals across all resolutions)
- Data: 5 environments, 37240 samples
- Learning rate: 0.01
- Number of epochs: 30

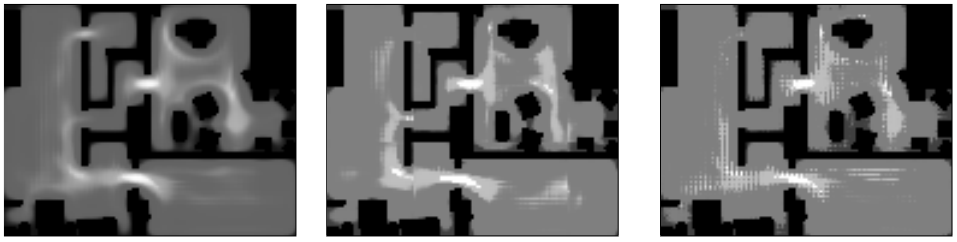
Algorithm 4.1: Map Annotation

Input: Map X of dimensions $r \times c$, h_θ with input $(n \times n)$
Output: Map annotated with h_θ

1: output $\leftarrow 0$
2: **for** Every $(n \times n)$ segment around $x \in X$ **do**
3: output$_x$ $\leftarrow$ output$_x$ $+h_\theta(x)$
4: **end for**
5: **if** EVALUATION **then**
6: output $\leftarrow$ quantize(output)
7: **else**
8: output $\leftarrow$ normalize(output)
9: **end if**
10: output $\leftarrow$ output $\circ X$
11: **return** output

4.3.3 Map annotation

After training h_θ, we use a sliding window approach to compute multiple predictions per pixel and avoid artifacts from predictions with insufficient topological context. Alg. 4.1 breaks down our proposed map annotation strategy. First, The map is iterated over with a stride of one and the model is applied to every possible $n \times n$ occupancy grid that contains a non-obstacle pixel (padding if necessary) (line 2). The predictions are accumulated in an array (line 3). For evaluation, the sum of predictions is then quantized such that the distribution of predictions matches a given distribution (line 6). The quantization subroutine



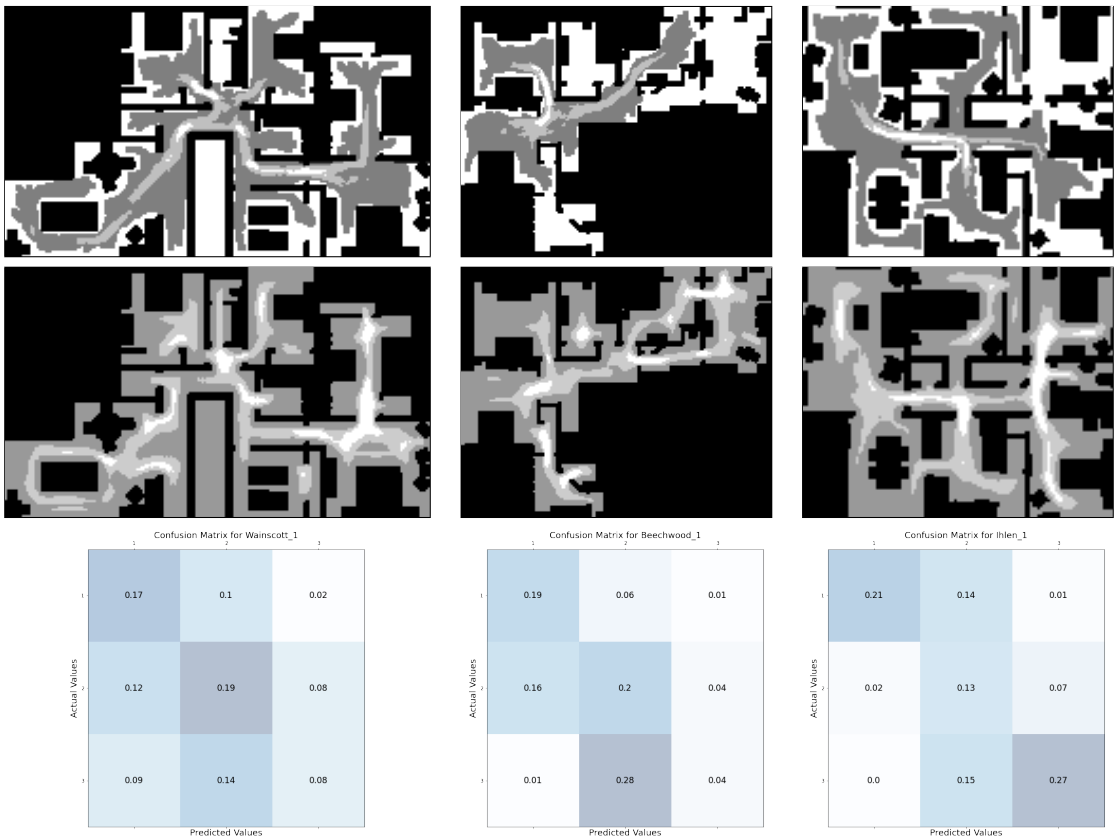
a) Proposed Strategy b) Large Stride c) Per Pixel Prediction

Figure 4.3: Ihlen 0 environment annotated with a) the proposed annotation strategy b) the strategy with a large stride/no overlaps, and c) a classification, prediction per pixel, strategy

finds the binning thresholds (i.e., percentiles) for each class and assigns the class label for each pixel accordingly. If a discrete output is not necessary or desired, the output is simply normalized (line 8). Finally, the predictions are multiplied element-wise with the input traversability map so that the annotated map contains the same obstacle geometry as the input (line 10). Fig. 4.3a depicts the output of our strategy. Fig. 4.3b shows an output with large stride and no overlap that contains artifacts from not averaging over predictions from multiple windows. There are noticeable differences around the boundary of the sliding window. This is because the window lacks the topological context that is beyond its boundary. On the other hand, Fig. 4.3c shows the output when the model only makes a single prediction per pixel during the sliding window annotation, practically approaching the annotation as classification rather than segmentation.

4.4 EXPERIMENTAL RESULTS

We use three testing environments for evaluation. As with the training data, the ground-truth labels for segmentation evaluation are noisy and imbalanced due to the randomly sampled starts and goals of the trajectories, the limited number of simulated trajectories used to generate the training data, and the sparse nature of criticality. While other metrics like betweenness centrality may be deterministic, we argue that this approach is more applicable to predicting the movement of agents controlled by an unknown policy or from real-world data. We opted against hand labeling the environments according to the guidelines described in Sec. 4.3.1 to prevent introducing any bias. For social cost prediction evaluation, we use the normalized annotations.

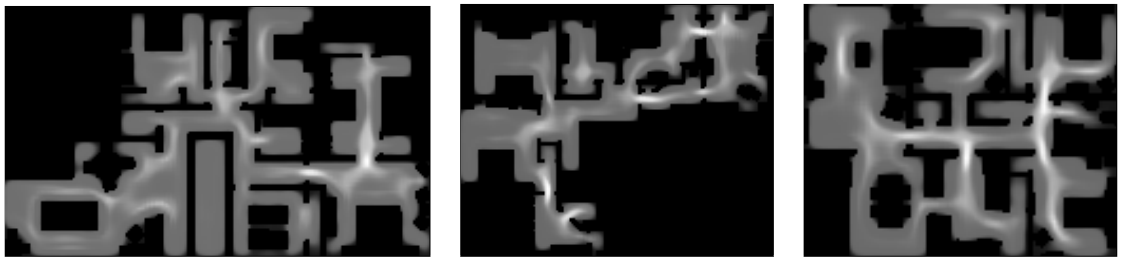


a) Wainscott 1 b) Beechwood 1 c) Ihlen 1

Figure 4.4: Ground-truth, predictions, and confusion matrices for Wainscott 1, Beechwood 1, and Ihlen 1 environments

4.4.1 Segmentation Evaluation

Test Environments Fig. 4.4 summarizes the results of our approach's segmentation predictions. The confusion matrices in Fig. 4.4c are scaled by the weights proposed in Sec. 4.3.2 and show that our predictions are generally correct with respect to the simulated data. Class value 1 corresponds to least frequently visited regions (darkest), while 2 and 3 correspond to frequently visited regions and the most frequently visited regions respectively. For Wainscott 1 and Ihlen 1, two of the predicted criticality values are mostly predicted correctly, while only one is mostly predicted correctly for Beechwood 1. Notably, whenever a prediction is wrong, it is almost always predicted as an adjacent criticality value. This is expected because of the noise in the ground-truth and reflects the nature of motion patterns. Qualitatively, the predictions in Fig. 4.4 capture the desired narrow passages and patterns. Overall, the correlation between the class labels and the predictions and our intuition of the desired output yield acceptable results.



a) Wainscott 1 b) Beechwood 1 c) Ihlen 1

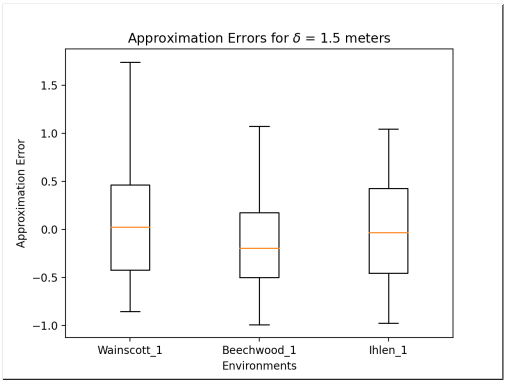
Figure 4.5: Test environments annotated with criticality predictions.

Unlabeled Regions There are additional opportunities to assess the performance of the network qualitatively. In over half of the training and testing environments, parts of the maps are not labeled in the simulated data. As seen in Figs. 4.4a-c, our model predicts the expected patterns and generalizes to unlabeled regions. This finding further motivates the use of machine learning for motion pattern prediction. While using the simulated training data or a non-learning-based method for pattern prediction may be more straightforward, the agent may not be able to accurately generalize the patterns to other rooms within the same environment.

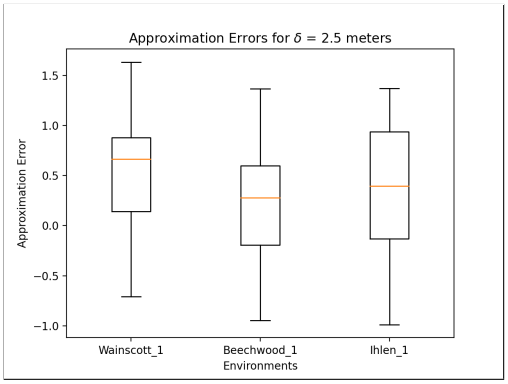
4.4.2 Social Cost Prediction Evaluation

Lastly, we evaluate whether the output of our approach is a good estimator of social costs as described in our motivation (Sec. 4.2). The goal is to approximate the social costs of a discrete trajectory by treating $h_\theta(x)$ as the probability of occupancy by a pedestrian in the annotated environments (See Fig. 4.5). We simulate 300 navigation episodes with more than four pedestrians each in the test environments (100 per environment) and compute the ground-truth and predicted social costs of one trajectory per navigation episode. To depict the comparison between the predicted costs and ground-truth, we measure the average approximation error at all timesteps during which the ground-truth social costs are not 0 (i.e., there is at least one pedestrian within δ meters).

In the experiments depicted in Fig. 4.6, we set σ to 1 meter and show the boxplots at two values of δ, 1.5 meters and 2.5 meters. Since our estimate always assumes that there is a non-zero probability of occupancy (as judged by the environment's topology), it inherently provides an incorrect estimate when no pedestrians are present. Generally, this stems from the stochastic nature of pedestrian trajectories and is the reason why we ignore timesteps with no pedestrians nearby. In other words, our predicted social cost is a conservative approximation since the robot may not encounter pedestrians during execution



a) Approximation errors for $\delta = 1.5$ meters



b) Approximation errors for $\delta = 2.5$ meters

Figure 4.6: Boxplots of approximation errors for 100 navigation episodes per environment.

time. The approximation errors have high variance across different episodes because they make the same assumptions regardless of the number of pedestrians within δ. Nonetheless, the median estimated error is close to 0 and over half of the approximation errors are smaller than 50%. When δ is increased, the approximations retain a similar distribution but shift to become almost entirely over-estimates. This is likely because the model is taking into account more positions and in a sense, assuming more pedestrians are likely to be nearby.

Overall, it is difficult to assess whether a model is a reasonable estimate for social costs due to the stochasticity of the problem. Despite this, the most crucial aspect of such a model would be its ability to estimate the social costs relative to other potential trajectories. In practice, robots can use incorrect estimates if the best trajectory has a lower estimate than the rest. As mentioned in Sec. 1 and is done in relevant related work, we intend to further evaluate the model by measuring its benefits to downstream applications in future work. Moreover, we note that, in practice, the robot would update its estimates as observations become available and that a robot could reasonably behave more cautiously than necessary in various situations when predicting social cost.

4.5 CONCLUSION

In this chapter, we proposed a novel methodology for learning to predict motion patterns from traversability maps through self-supervision, emphasizing its application in dynamic environments. Central to our approach is the utilization of crowd simulation data from real-world apartments. This marks a departure from traditional multi-agent and sampling-based motion planning methods, shifting the focus to a more comprehensive understanding of long-term pedestrian motion. Our methodology not only enhances the ability of agents to

navigate shared spaces but also contributes to a deeper understanding of social navigation dynamics, particularly in predicting pedestrian movement for preemptive collision avoidance and social cost reduction.

A key contribution of this chapter is demonstrating our model's capability to generalize across different environments, illustrating its adaptability and relevance in various settings. The model's effectiveness in identifying regions of high pedestrian traffic and predicting long-horizon motion patterns underlines its potential to improve social navigation solutions. This aspect is particularly crucial in environments where traditional navigation systems might struggle, such as crowded urban areas or complex indoor spaces. By successfully extending our approach to new environments, we've shown that our model can be a robust tool for enhancing the interaction dynamics between robots and humans in shared spaces.

The analysis of social cost prediction forms an integral part of our research. Our model can reasonably approximate these costs by treating occupancy probability as a critical factor in evaluating social interactions. Despite the inherent challenges posed by the stochastic nature of pedestrian trajectories, our approach offers a conservative yet effective estimation of social costs. This capability is vital in ensuring safe and socially aware navigation, making our model a valuable asset in social robotics. Overall, this chapter highlights the significant strides we've made in advancing motion pattern prediction and its application in social navigation.

CHAPTER 5: CONCLUSIONS AND FUTURE WORK

5.1 CONCLUSION

This thesis has covered a large swath of the field of robotics, mainly focusing on the challenges of navigating dynamic environments and multi-agent motion planning (MAMP). Our research began with a focus on sampling-based motion planning for multi-agent systems, where we addressed the intricacies of computing feasible paths amidst multiple agents efficiently. The concept of navigation among multiple agents in constrained environments was particularly emphasized, given its difficulty and significance in deploying robots in the real world.

In tackling the challenges of MAMP, we proposed Avoidance Critical Probabilistic Roadmaps (ACPRMs). This innovation, presented in Chapter 3, marked a significant innovation in sampling-based MAMP. ACPRMs emerged as a solution to the limitations of standard sampling strategies in multi-agent scenarios. We proposed an algorithm for constructing sparse roadmaps by identifying regions critical to MAMP and integrating concepts like *betweenness centrality*, facilitating efficient navigation in constrained spaces. This methodology advanced the technical aspects of motion planning and enhanced robot interaction dynamics in shared spaces.

The latter part of our research, detailed in Chapter 4, shifted focus towards a broader application. By leveraging crowd simulation data from real-world apartments, we expanded our methodology to include predicting long-term motion patterns in dynamic environments more formally. This shift marks a notable step towards enhancing social navigation solutions, especially in predicting pedestrian movement for proactive collision avoidance and social cost minimization. Our model's adaptability and effectiveness in identifying high pedestrian traffic areas and predicting long-horizon motion patterns showcased its potential to improve social navigation, particularly in complex urban or indoor spaces.

In summary, this thesis substantially contributes to MAMP and social navigation. Our work has laid a foundation for future research, highlighting the importance of predicting long-horizon pedestrian behavior and the need for models that can adapt to various dynamic environments. The journey through this research has provided valuable insights and advancements in the technical aspects of robotics and opened new avenues for enhancing the coexistence and cooperation between robots and humans in shared spaces. The path forward is clear: to continue exploring, refining, and innovating these types of approaches, with the ultimate goal of integrating robots into our dynamic world.

5.2 LIMITATIONS

Reflecting on the entirety of this thesis, it is apparent that while our research has made significant strides in advancing MAMP and social navigation, there are inherent limitations and open questions that merit acknowledgment and future exploration.

One primary limitation pertains to the stochastic nature of dynamic environments and the novelty of our approach. The unpredictable movement patterns of agents, especially in highly dynamic settings, pose a challenge to the development and evaluation of our models. Ideally, our work would align with universally accepted ground truths and established metrics for social navigation. However, the unique challenges of our problem space necessitated the creation of new evaluation techniques. Consequently, the effectiveness and applicability of our approach are most pronounced in specific scenarios characterized by high pedestrian density or the necessity for agents to navigate through limited spatial regions. While these scenarios are not uncommon in real-world settings, our methods' broader applicability and utility are more nuanced and context-dependent and, therefore, need more evaluation.

Another critical limitation arises from our reliance on simulated data. Although the simulation-to-real-world transition is theoretically robust—given that our model does not depend on physics simulation or high-dimensional sensor inputs like lidar scans—the learned motion patterns are reflective of simulated pedestrian behavior rather than actual human movement. A dataset encompassing real-world pedestrian behavior would undoubtedly provide richer semantic information and enhance the model's practical deployment. However, assembling such a comprehensive dataset without relying on simulation presents substantial challenges. It would necessitate extensive data collection efforts, likely involving cameras and other sensors, without guaranteed coverage of the entire environment. At present, no real-world dataset exists that offers the depth and breadth required for validating our approach. The development of such datasets, alongside more advanced simulators, would significantly contribute to grounding our work.

5.3 FUTURE WORK

Building upon the foundation laid by this thesis, several promising avenues for future research could significantly enhance the field of MAMP and social navigation. These future directions aim to address the identified limitations and push the boundaries of our current understanding and capabilities.

Real-World Data Integration and Validation A critical step forward would be integrating real-world pedestrian data into our models. This integration would help validate the effectiveness of our methodologies and provide more nuanced insights into human behavior and movement patterns. Future research should focus on developing methods to collect and incorporate such data, possibly through crowd-sourced data collection methods.

Improved Simulation Models To bridge the gap between simulation and reality, future work should also aim at improving the fidelity of simulation models. This improvement involves incorporating more realistic pedestrian behaviors and environmental interactions into the simulation frameworks. By enhancing the realism of simulated scenarios, we can better prepare our models for the complexities and unpredictability of real-world environments.

Adaptation to Diverse Environmental Contexts Expanding the application of our methodologies to a broader range of environments, including outdoor urban settings, industrial settings, or complex indoor environments like hospitals, would be a valuable direction. Research in this area would involve adapting our models to more diverse spatial layouts and dynamic conditions, ensuring robustness in varying contexts.

In conclusion, the roadmap for future research both challenging and exciting. It promises to yield significant advancements in how robots navigate and interact within dynamic, human-centric environments, ultimately leading to better coexistence between robots and humans.

REFERENCES

[1] S. M. LaValle, *Planning Algorithms*. Cambridge, U.K.: Cambridge University Press, 2006, available at http://planning.cs.uiuc.edu/.

[2] L. E. Kavraki, P. Švestka, J. C. Latombe, and M. H. Overmars, "Probabilistic roadmaps for path planning in high-dimensional configuration spaces," *IEEE Trans. Robot. Automat.*, vol. 12, no. 4, pp. 566–580, August 1996.

[3] K. Kondo, "Motion planning with six degrees of freedom by multistrategic bidirectional heuristic free space enumeration," *IEEE Trans. Robot. Automat.*, vol. 7, no. 3, pp. 267–277, 1991.

[4] B. Ichter, E. Schmerling, T.-W. E. Lee, and A. Faust, "Learned critical probabilistic roadmaps for robotic motion planning," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2020, pp. 9535–9541.

[5] L. C. Freeman, "A set of measures of centrality based on betweenness," *Sociometry*, pp. 35–41, 1977.

[6] C. Pérez-D'Arpino, C. Liu, P. Goebel, R. Martín-Martín, and S. Savarese, "Robot navigation in constrained pedestrian environments using reinforcement learning," *arXiv preprint arXiv:2010.08600*, 2020.

[7] C. Fulgenzi, A. Spalanzani, and C. Laugier, "Dynamic obstacle avoidance in uncertain environment combining pvos and occupancy grid," in *Proceedings 2007 IEEE International Conference on Robotics and Automation*. IEEE, 2007, pp. 1610–1616.

[8] F. F. Arias, B. Ichter, A. Faust, and N. M. Amato, "Avoidance critical probabilistic roadmaps for motion planning in dynamic environments," in *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 10 264–10 270.

[9] G. Sharon, R. Stern, A. Felner, and N. R. Sturtevant, "Conflict-based search for optimal multi-agent pathfinding," *Artificial Intelligence*, vol. 219, pp. 40–66, 2015.

[10] M. Bennewitz, W. Burgard, and S. Thrun, "Optimizing schedules for prioritized path planning of multi-robot systems," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2001, pp. 271–276.

[11] J. Abbenseth, F. G. Lopez, C. Henkel, and S. Dörr, "Cloud-based cooperative navigation for mobile service robots in dynamic industrial environments," in *Proceedings of the Symposium on Applied Computing*, 2017, pp. 283–288.

[12] J. van den Berg, M. Lin, and D. Manocha, "Reciprocal velocity obstacles for real-time multi-agent navigation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*. IEEE, 2008, pp. 1928–1935.

[13] M. Phillips and M. Likhachev, "Sipp: Safe interval path planning for dynamic environments," in *2011 IEEE International Conference on Robotics and Automation*. IEEE, 2011, pp. 5628–5635.

[14] I. Solis, R. Sandström, J. Motes, and N. M. Amato, "Roadmap-optimal multi-robot motion planning using conflict-based search," *arXiv preprint arXiv:1909.13352*, 2019.

[15] B. Brüggemann and D. Schulz, "Coordinated navigation of multi-robot systems with binary constraints," in *2010 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2010, pp. 3854–3859.

[16] M. Gharbi, J. Cortés, and T. Siméon, "Roadmap composition for multi-arm systems path planning," in *2009 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2009, pp. 2471–2476.

[17] Z. Yao and K. Gupta, "Distributed roadmaps for robot navigation in sensor networks," *IEEE Transactions on Robotics*, vol. 27, no. 5, pp. 997–1004, 2011.

[18] S. Kumar and S. Chakravorty, "Multi-agent generalized probabilistic roadmaps: Mag-prm," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2012, pp. 3747–3753.

[19] C. Henkel and M. Toussaint, "Optimized directed roadmap graph for multi-agent path finding using stochastic gradient descent," in *Proceedings of the 35th Annual ACM Symposium on Applied Computing*, 2020, pp. 776–783.

[20] N. M. Amato, O. B. Bayazit, L. K. Dale, C. Jones, and D. Vallejo, "OBPRM: an obstacle-based PRM for 3d workspaces," in *Proceedings of the third Workshop on the Algorithmic Foundations of Robotics*. Natick, MA, USA: A. K. Peters, Ltd., 1998, (WAFR '98). pp. 155–168.

[21] D. Hsu, L. E. Kavraki, J.-C. Latombe, R. Motwani, and S. Sorkin, "On finding narrow passages with probabilistic roadmap planners," in *Proc. Int. Workshop on Algorithmic Foundations of Robotics (WAFR)*, 1998, pp. 141–153.

[22] D. Hsu, G. Sánchez-Ante, and Z. Sun, "Hybrid PRM sampling with a cost-sensitive adaptive strategy," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*. Barcelona, Spain: IEEE, 2005, pp. 3874–3880.

[23] H. Kurniawati and D. Hsu, "Workspace importance sampling for probabilistic roadmap planning," in *Proc. IEEE Int. Conf. Intel. Rob. Syst. (IROS)*, vol. 2, sept.-2 oct. 2004, pp. 1618–1623.

[24] J. Berg and M. Overmars, "Using workspace information as a guide to non-uniform sampling in probabilistic roadmap planners," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2004, pp. 453–460.

[25] B. Ichter, J. Harrison, and M. Pavone, "Learning sampling distributions for robot motion planning," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2018, pp. 7087–7094.

[26] M. Zucker, J. Kuffner, and J. A. Bagnell, "Adaptive workspace biasing for sampling-based planners," in *2008 IEEE International Conference on Robotics and Automation*. IEEE, 2008, pp. 3757–3762.

[27] C. Zhang, J. Huh, and D. D. Lee, "Learning implicit sampling distributions for motion planning," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2018, pp. 3654–3661.

[28] P. Trautman and A. Krause, "Unfreezing the robot: Navigation in dense, interacting crowds," in *2010 IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2010, pp. 797–803.

[29] P. Trautman, J. Ma, R. M. Murray, and A. Krause, "Robot navigation in dense human crowds: Statistical models and experimental studies of human–robot cooperation," *The International Journal of Robotics Research*, vol. 34, no. 3, pp. 335–356, 2015.

[30] U. Muller, J. Ben, E. Cosatto, B. Flepp, and Y. Cun, "Off-road obstacle avoidance through end-to-end learning," *Advances in neural information processing systems*, vol. 18, 2005.

[31] M. Bojarski, D. Del Testa, D. Dworakowski, B. Firner, B. Flepp, P. Goyal, L. D. Jackel, M. Monfort, U. Muller, J. Zhang et al., "End to end learning for self-driving cars," *arXiv preprint arXiv:1604.07316*, 2016.

[32] A. Pokle, R. Martín-Martín, P. Goebel, V. Chow, H. M. Ewald, J. Yang, Z. Wang, A. Sadeghian, D. Sadigh, S. Savarese et al., "Deep local trajectory replanning and control for robot navigation," in *2019 International Conference on Robotics and Automation (ICRA)*. IEEE, 2019, pp. 5815–5822.

[33] Y. F. Chen, M. Everett, M. Liu, and J. P. How, "Socially aware motion planning with deep reinforcement learning," in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2017, pp. 1343–1350.

[34] C. Chen, Y. Liu, S. Kreiss, and A. Alahi, "Crowd-robot interaction: Crowd-aware robot navigation with attention-based deep reinforcement learning," in *2019 International Conference on Robotics and Automation (ICRA)*. IEEE, 2019, pp. 6015–6022.

[35] D. Ellis, E. Sommerlade, and I. Reid, "Modelling pedestrian trajectory patterns with gaussian processes," in *2009 IEEE 12th International Conference on Computer Vision Workshops, ICCV Workshops*. IEEE, 2009, pp. 1229–1234.

[36] Y. Yoo, K. Yun, S. Yun, J. Hong, H. Jeong, and J. Young Choi, "Visual path prediction in complex scenes with crowded moving objects," in *Proceedings of the ieee conference on computer vision and pattern recognition*, 2016, pp. 2668–2677.

[37] K. Kim, D. Lee, and I. Essa, "Gaussian process regression flow for analysis of motion trajectories," in *2011 International Conference on Computer Vision*. IEEE, 2011, pp. 1164–1171.

[38] M. K. C. Tay and C. Laugier, "Modelling smooth paths using gaussian processes," in *Field and Service Robotics*. Springer, 2008, pp. 381–390.

[39] A. Rudenko, L. Palmieri, M. Herman, K. M. Kitani, D. M. Gavrila, and K. O. Arras, "Human motion trajectory prediction: A survey," *The International Journal of Robotics Research*, vol. 39, no. 8, pp. 895–935, 2020.

[40] E. A. Pool, J. F. Kooij, and D. M. Gavrila, "Using road topology to improve cyclist path prediction," in *2017 IEEE Intelligent Vehicles Symposium (IV)*. IEEE, 2017, pp. 289–296.

[41] A. Rudenko, L. Palmieri, and K. O. Arras, "Predictive planning for a mobile robot in human environments," in *Proc. of the IEEE Int. Conf. on Robotics and Automation (ICRA), Works. on AI Planning and Robotics*, 2017.

[42] A. Rudenko, L. Palmieri, A. J. Lilienthal, and K. O. Arras, "Human motion prediction under social grouping constraints," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2018, pp. 3358–3364.

[43] J. F. Kooij, F. Flohr, E. A. Pool, and D. M. Gavrila, "Context-based path prediction for targets with switching dynamics," *International Journal of Computer Vision*, vol. 127, no. 3, pp. 239–262, 2019.

[44] S. Minaee, Y. Y. Boykov, F. Porikli, A. J. Plaza, N. Kehtarnavaz, and D. Terzopoulos, "Image segmentation using deep learning: A survey," *IEEE transactions on pattern analysis and machine intelligence*, 2021.

[45] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 3431–3440.

[46] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *International Conference on Medical image computing and computer-assisted intervention*. Springer, 2015, pp. 234–241.

[47] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.

[48] Drow, "Random dungeon generator," Donjon, http://donjon.bin.sh/.

[49] S. A. Wilmarth, N. M. Amato, and P. F. Stiller, "MAPRM: A probabilistic roadmap planner with sampling on the medial axis of the free space," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, vol. 2, 1999, pp. 1024–1031.

[50] F. Xia, A. R. Zamir, Z. He, A. Sax, J. Malik, and S. Savarese, "Gibson env: Real-world perception for embodied agents," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 9068–9079.

[51] C. Mavrogiannis, F. Baldini, A. Wang, D. Zhao, P. Trautman, A. Steinfeld, and J. Oh, "Core challenges of social robot navigation: A survey," *arXiv preprint arXiv:2103.05668*, 2021.

[52] T. Kruse, A. K. Pandey, R. Alami, and A. Kirsch, "Human-aware robot navigation: A survey," *Robotics and Autonomous Systems*, vol. 61, no. 12, pp. 1726–1743, 2013.

[53] J. van den Berg, S. J. Guy, J. Snape, M. C. Lin, and D. Manocha, "Rvo2 library: Reciprocal collision avoidance for real-time multi-agent simulation," 2011.

[54] F. Xia, W. B. Shen, C. Li, P. Kasimbeg, M. E. Tchapmi, A. Toshev, R. Martín-Martín, and S. Savarese, "Interactive gibson benchmark: A benchmark for interactive navigation in cluttered environments," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 713–720, 2020.

[55] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.